

Confidential

产品开发更改

DC No. / 开发更改编号: RBCE-PDECR2026003

Date / 日期: 2026.04.02

Affected domain / 影响的开发域 ☐ SYS ☒ ME ☐ HW

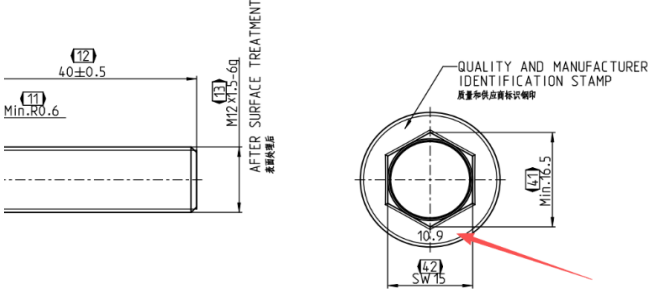
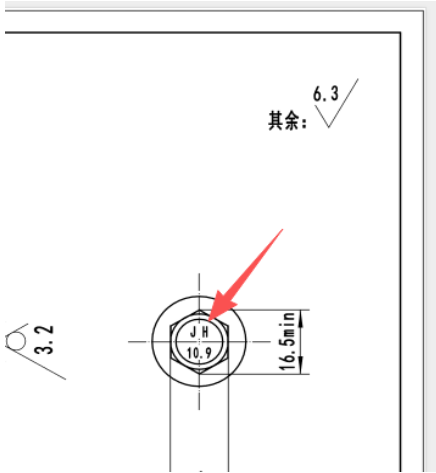
Step 1: Change request & Basic information

Customer project Name/ 客户项目名称:	JP360	MCR No. / MCR号:	
Affect Product No. / 产品号:	F03ZS0000B-04	Component No. / 零部件号:	F03Z20088M-00
Sample type / 样品类型:	☐ A ☐ B ☐ C ☒ FD	Initiator/发起人:	lu feihong
Reason of changes / 更改理由:			

螺栓增加二供供应商, 新增二供螺栓料号及图纸, F03Z20088M-00, 该二供料号图纸与一供图纸区别为,质量和供应商标识区域调整

Customer request / 客户要求 ☐	Supplier request / 供应商请求 ☒	Design optimization / 设计优化 ☒
Correction / 更正(错误) ☐	Process / 工艺 ☐	Other / 其它 ☐

Step 2: Change proposal & Change discription

当前设计	更改建议	备注
		新增螺栓供应商, 新螺栓图纸调整二供供应商质量和供应商标识, 其余尺寸未作更改
当前设计是标识打在螺栓法兰处	更改后标识打在螺栓头部	

 BOSCH	Impact analysis		Edition 2.0		
Confidential					
Corresponding DC No. / 对应的开发更改编号: RBCE-PDECR2026003					
Date / 日期: 2026.03.17					
Step 3.1 Impact analysis / 影响分析					
1. Function&Performance will be influenced? / 产品功能性能影响?		☒ no/否	☐ yes/是	Confirmed by: _lu feihong	
2. Interface and Apperance will be influenced? / 接口和外观影响?		☒ no/否	☐ yes/是	Confirmed by: __lu feihong__	
3. Reliability and robustness will be influenced? / 产品可靠性、鲁棒性影响?		☐ no/否	☒ yes/是		
Description of above points, and concern point analysis / 对上述影响的描述, 以及分析是否存在顾虑点					
螺栓疲劳试验未完成, 带测试验证					
4. Others component will be influenced? / 影响其它零部件?		☒ no/否	☐ yes/是	Confirmed by: _huangwei_____	
If yes, Which components will change in parallel? 如果影响, 哪些零部件也需要同时变更?					
5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?		☐ no/否	☒ yes/是	Confirmed by: __Hu Bin_____	
1.确认产线螺栓拧紧工艺, 一供螺栓F03Z20046A-00与二供F03Z20088M-00拧紧工艺相同					
6. Influence on supplier part? Feedback from Supplier/Purchasing?		☐ yes/是	☒ no/否	Confirmed by: ____yang wei_____	
采购件变更					
7. Influence on System/Hardware/Software/Calibration/Mechanical parts?		☐ yes/是	☒ no/否	Confirmed by: ____lu feihong_____	
e.g.: If changes influnce software/calibration, and ME/HW version alignment is required. e.g. SW reflashing affected products in stock e.g. Mechanical change or HW change triggered SW change, inform SW team e.g. System element change trigger calibration change, inform Calibration team					
8. Influence on cost / 对成本的影响		☐ increase/增加	☒ decrease/降低	☐ no change/不变	
2.52CNY/ Motor= 14pcs (0.18CNY/ pc)					
9. Stock/Delivery treatment / 库存(发货)处理指示: 可以使用					
Mixed Deliveries Permissible? 改前改后是否可以混合供货? ☒ YES ☐ NO 单电机不混, 整托可混					
How to deal with 1st delivery after change? 改后第一批货物的发货要求? Answer回答: ____维持跟先前一样_____					
Raw materials: ☒ Not affect ☐ Use in other products ☐ Scrap ☐ Rework ☐ Use up (包含在途)					
Parts/Subassemblies: ☒ Not affect ☐ Use in other products ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(inhouse): ☒ Not affect ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(RDC外库): ☒ Not affect ☐ Scrap ☐ Rework ☐ Use up					
Finished goods(customer): ☒ Not affect ☐ Recall ☐ Rework (包含在途)					
Step 3.2 Quality Assurance Items / 检查项目					
Validations 验证		Plan finish date 计划完成日期	Resp.person 负责人	Comments 备注	
☒ try run		20260407	hu bin	一供与二供螺栓拧紧工艺共用, 验证	
☒ Capability Studies_ CMK		20260407	hu bin	一供与二供螺栓拧紧工艺共用, 验证	
☐ Capability Studies_ MSA					
☐ MAE release					
☐ Cleanness test					
☐ QZ test					
☐ 200h PDL					
☒ BOM check		20260331	lu feihong	双供, 均适用	
☐ Test report					
☐ PAV release					
螺栓盐雾试验		20260331	yang wei		
螺栓疲劳测试		20260530	yang wei		
Step 3.3 Affected documents Check/ 影响文档检查					
1.Interface FMEA-relevant / IFMEA:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
2.Product FMEA-relevant / DFMEA:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
3.Special Charecteristics relevant/PSC:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
4.IMDS relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
5.Offer drawing relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
6.TCD relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
7.Norm, WB, HF... relevant:		☒ no/否 ☐ yes/是	Resp.person:	Due date: _____	
8.WI check		☐ no/否 ☒ yes/是	Resp.person: _____xie qiangqiang	Due date: ____ 20260331	
Step 4 Technical feasibility & Validation plan approval / 技术可行性&验证计划批准					
Development/ 研发	Purchasing/采购	TEF/工艺	Quality/质量	CPjM/客户项目	COS/样品
xiang liangshan	tao jiong	he yonggang	xia qian	su jian	he yonggang
MOE/ 生产	LOG/物流	Others (in needed)			
luo zhi	xu baochun				
* Note: if reflash SW in joint location, e.g. in ME plant, the max. 2 resp. are needed to approve the PD-ECR, e.g. MOE DM +EPQ					

Example of affected actions

Related domains
ME
Specific analysis points

HW
Specific analysis points

SW
Specific analysis points

SW
Specific analysis points

SW
Specific analysis points

PD-ECR check points

Step 3.1 Impact analysis / 影响分析

5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?

- whether production fixture needs to consider Poka Yoke to prevent manual assembly with wrong direction in Types change?
- whether change over check list needs to prepare for operators when type changes?
- whether the PLC program and working parameter for Screw tightening needs updating?
- whether additional assembly/testing risks by the change?

Step 3.2 Quality Assurance Items / 检查项目

- BOM check
- Test report
- PAV release

Step 3.1 Impact analysis / 影响分析

5. Manufactory/assembly/testing will be influenced? / 加工, 装配, 测试是否影响?

SW reflash operation needs to do:

- Scenario 1 (Highest efficient way): Rework at Customer side, reflash SW at EOL station or by OTA.
- Scenario 2: Return to Bosch Plant: Rework at internal area for SW reflash, key points to be defined: Rework site, Operator, ReflashTool, necessary Documents, RASIC

Step 6.1 Implementation check list / 导入清单

SW introduction to Production:

- RBCD(Calibration engineer) offline deliver updated SW code by mail to ME plant for try run
- ME upload to manufacturing SW flash station (interface?)
- ME try run with the new upgrated SW and feedback result to RBCD
- officially send upgrated SW to ME via workon during PD-ECR approval (2days)
(update SW if need based on the try run result)

Step 6.1 Implementation check list / 导入清单

e.g. Machanic electrical valvel design changed, triggered SW version updated, Prevention defined to prevent from Label/type error:

add SW identification code into Barcode for ensuring traceability



Team discussion results to be documented

Nr.	Change content	version	Date
1	Initial version	v1	Apr. 2025
2	1) Change scope cover SYS,ME, HW 2) Add linkage of SW activity	v2	Dec. 2025

Editor

Shi Kai (RBCD/EFC2)

Shi Kai (RBCD/EFC2)

Fan Xiaofeng (RBCD/ETC6)